

Mixed Reality and Digital Twin Interface for Robot Collaborative Tasks

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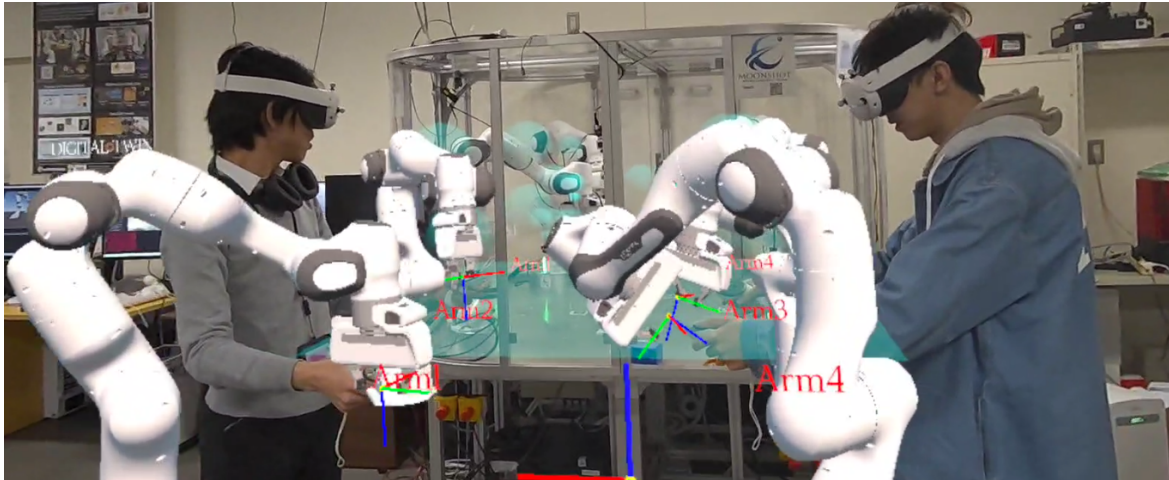


Figure 1: Interactive operation of the Mixed Reality Digital Twin and robot system by two users.

ABSTRACT

Traditional bimanual user interfaces and real-time video feeds are commonly used for robot control but become increasingly complex when managing multiple users and robots in collaborative tasks. Designing intuitive interfaces for multi-arm robotic systems is particularly challenging, especially in scenarios where direct in-situ control is impractical due to biohazards or stringent sterilization requirements. A Digital Twin (DT) offers a promising solution for monitoring and enabling interactive control of robotic systems in such environments. This work introduces a Mixed Reality (MR) interface that leverages the DT to provide operators with an intuitive and immersive means of controlling multi-arm robotic systems, enhancing efficiency and collaboration in complex robotic tasks.

Index Terms: Mixed / augmented reality, Collaborative interaction, Virtual reality

1 INTRODUCTION

The control of robotic systems has traditionally relied on bimanual user interfaces and real-time video feeds, which are effective for simple operations. However, as tasks grow more complex, particularly in collaborative scenarios involving multiple users and robotic units, these traditional approaches become increasingly challenging to manage. Designing intuitive interfaces for such multi-arm robotic systems is especially difficult when precise coordination and interaction are required. This is compounded by environments where direct, in-situ control is not feasible due to limitations like biohazards or sterilization requirements.

In these situations, virtual reality (VR) simulators provide a valuable alternative by offering a safe and accessible testing, planning, and training platform. These simulators replicate robotic behavior and operational conditions in a fully virtual environment, making them particularly useful in scenarios where physical operation is prohibitively costly, dangerous, or impractical. Despite their advantages, standalone VR simulators are limited by their inability to reflect the real-time state of the physical system.

Digital Twins (DT) address this limitation by extending the functionality of VR simulators. By integrating real-time data from the physical robot, DT creates a dynamic connection between the virtual and physical environments. Unlike static simulations, DT continuously updates based on sensor data and the robot's actual state, enabling real-time monitoring and interaction. This real-time synchronization makes DT indispensable for applications requiring situational awareness, diagnostics, and informed decision-making, bridging the gap between simulation and real-world operations.

Building on the capabilities of DT, Mixed Reality (MR) offers an even more immersive interface by seamlessly blending real and virtual environments. MR enhances the user experience through augmented reality (AR), which overlays critical visual information onto the physical world, and virtual reality (VR), which enables immersive simulations. Together, these technologies allow operators to interact intuitively with both physical and virtual robots, especially in scenarios where direct manipulation or access is limited. MR further improves situational awareness and operational safety, providing a versatile tool for robotic control in complex environments.

1.1 Related work

MR has been widely proposed in the literature as a human-machine interface for interactive and more intuitive control of robots. Mourtzis et al. [5] proposed a framework for human-robot interface utilizing MR as the human-machine interface for remote and safe operation of the robot. They used a DT of the robotic arm for AR visualization of safety zones. The system was evaluated in assembly tasks in the context of laboratory-based machine shop, al-

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though it was limited to single-user and single-robot operation. Ostannin et al. [6] introduced a framework for controlling an industrial robot arm using a MR interface, providing intuitive visual feedback, such as robot path visualization, and leveraging point cloud data to align the virtual simulation with the physical robot setup. Khatib et al. [3] proposed an MR interface for safe human-robot collaboration in manufacturing processes, utilizing RGB-D sensors to identify static and dynamic obstacles that the robot should avoid based on distance computations. Szczurek et al. [7] developed an MR human-robot interface designed for remote operations in hazardous environments. This multi-modal system integrates hand, eye, and motion tracking, as well as voice recognition for control, while offering video, 3D point cloud, and audio feedback from the robot. The system also enables collaboration among multiple experts, whether in the same location or operating remotely. MR interfaces have also been explored for outdoor mobile robots. Chen et al. [1] presented an MR interface for real-time multi-agent visualization, which leverages visual SLAM for environment mapping and aligns the robots and MR devices to a common reference frame, while employing hand gestures to control the robots. MR human-robot interfaces are useful not only for teleoperation but also for training AI models. El Hafi et al. [2] proposed a teaching system for multimodal object categorization through human-robot interaction via MR visualization, aimed at addressing unexpected results and accelerating the learning process of service robots.

1.2 Statement of contributions

This work proposes a MR interface integrated with a DT to provide intuitive and immersive solution for controlling multi-arm robotic systems as shown in Fig. 1. The approach is designed to enhance user interaction and improve efficiency, particularly in complex collaborative robotic tasks. The process of developing this MR human-robot interface and preliminary trials of the system is reported.

2 MATERIAL AND METHODS

The system comprises a DT that replicates the state of the robot system and other objects in the scene, such as the pose of each head-mounted display (HMD), and serves as the central server. Clients connect to the DT server via TCP, including the HMD (Meta Quest 3), which acts as the user interface for each operator, and the robot system controller. To unify the coordinate system, the OptiTrack (NaturalPoint, USA) motion tracking system is used to track the poses of the headsets and the robot workspace relative to an absolute origin. Each subsystem and other relevant components are briefly described in the following subsections.

2.1 Robot system

The robot system shown in Fig. 2 consists of four Franka Research 3 (Franka Robotics, Germany) robot arms with 7 degrees of freedom (7-DOF), each equipped with a Franka Hand (Franka Robotics, Germany) as the end effector. These arms are mounted on a rigid frame, forming a multi-arm robotic system designed for autonomous collaborative manipulation in scientific exploration. The frame housing the robots is enclosed with transparent panels to simulate the experimental conditions requiring sterilization. To ensure safe operation with joint limit and collision constraints, a custom software based on the vector field inequality (VFI) framework was implemented [4] to allow multi-robot interaction. This framework allows for the definition of simple primitives (point, plane, line, or finite line) that are kinematically attached to the robot arm. These geometries serve as mutual interpenetration constraints, augmenting the IK task for collision avoidance.

2.2 Digital Twin

The DT scene was constructed using the CAD models employed to design the robot system frame and the CAD models of the robot



Figure 2: Robot platform for scientific exploration with four robot arms.

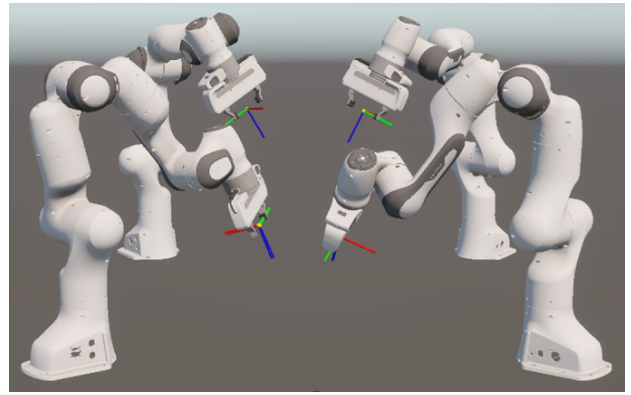


Figure 3: A screenshot of the DT mirroring the real robot system.

arm provided by the manufacturer. This ensures an accurate virtual replica of the physical system in a virtual environment. The DT application was developed using the Unity 3D engine, for real-time rendering of the virtual scene (see Fig. 3). The DT includes a server program that manages the scene state variables and handles client queries to retrieve or update these variables.

2.3 User interface

A commercial HMD, the Meta Quest 3 (Meta, USA), along with its handheld game controllers, is used as a user interface bridging the operator in the physical world with the DT in the virtual space. The HMD features built-in cameras that enable a see-through mode, supporting a wide range of MR applications, and utilizes inside-out tracking through onboard cameras and sensors to determine its relative position and orientation. However, it lacks built-in absolute motion tracking within a room, which is necessary for precise synchronization across multiple devices or users in a shared physical space.

2.4 Motion tracking

The OptiTrack motion tracking system is utilized to establish a unified coordinate system for all devices within the environment. This system employs infrared (IR) cameras and IR reflective markers, along with dedicated tracking software Motive, to track and estimate the poses of objects. Calibration of the motion tracking system is performed to define an absolute origin inside the room. To track the pose of HMDs, five 12.7 mm IR reflective markers are attached to each HMD. The Motive software assigns unique IDs to

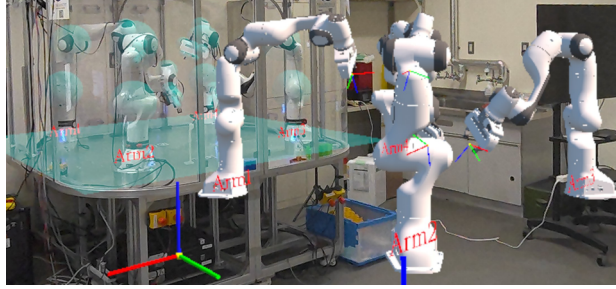


Figure 4: The MR application as seen from the HMD, the physical robot on the background with an overlay of the collision avoidance constraints, and the DT in the front.

these marker sets and estimates a rigid body pose for each HMD. However, the estimated pose does not align with the true reference frame of the HMD, defined somewhere between the eyes, necessitating additional calibration to resolve this discrepancy. This calibration is done for each HMD only once and it remains fixed as long as the object definition in the OptiTrack application is unchanged. This calibration ensures that all HMDs share the same absolute origin inside the room.

2.5 MR application

The MR application, compatible with the Meta Quest 3 HMD, was developed using Unity3D and the OpenXR framework. The virtual scene includes a replica of the DT, with four robot arms mirroring the configuration of the real robot arms. The reference frame of the DT is distinct from that of the real robot, allowing the user to interact with the DT without needing to approach the physical robot. This setup is particularly suitable for simulating hazardous or sterile environments. Translucent collision shapes are overlaid in Augmented Reality on the real robot, providing the operator with visual cues regarding collision constraints (See Fig. 4).

The interaction begins when the user wears the HMD and launches the application. Pressing the homing button on the game controller sets the reference frame, aligning the DT with the real robot and synchronizing with the robot controller. White arrows act as hand cursors, positioned in 3D space according to the user's hand movements (see Fig. 5-top). To interact with the robot, the user moves the cursor toward the DT's end effector, where expanding axis vectors indicate readiness for interaction. Pressing the trigger button gives control of the robot arm, updating its target pose based on the user's hand movements (see Fig. 5-bottom). An additional button controls the gripper, and releasing the trigger button stops the teleoperation, leaving the robot arm stationary.

3 RESULTS AND DISCUSSION

The proposed DT and MR user interface underwent preliminary testing for interactive operation of a robotic system with four robot arms, allowing up to four operators to work simultaneously. Although the DT was positioned near the physical robot system during this test, it is designed for remote deployment in practical applications. The tests demonstrated that the DT accurately mirrored the physical robot system in real-time. The communication protocol provided fast and reliable data exchange among four HMDs, the DT server, and the robot controller, with no perceivable latency. Following the calibration of the HMDs to a unified coordinate system, minimal alignment errors were observed between the physical robots and the AR overlays. The unified coordinate system ensured that all users shared an identical virtual scene, preventing collisions among human operators working in the same space while wearing HMDs. This feature is critical for ensuring the safety of the system,

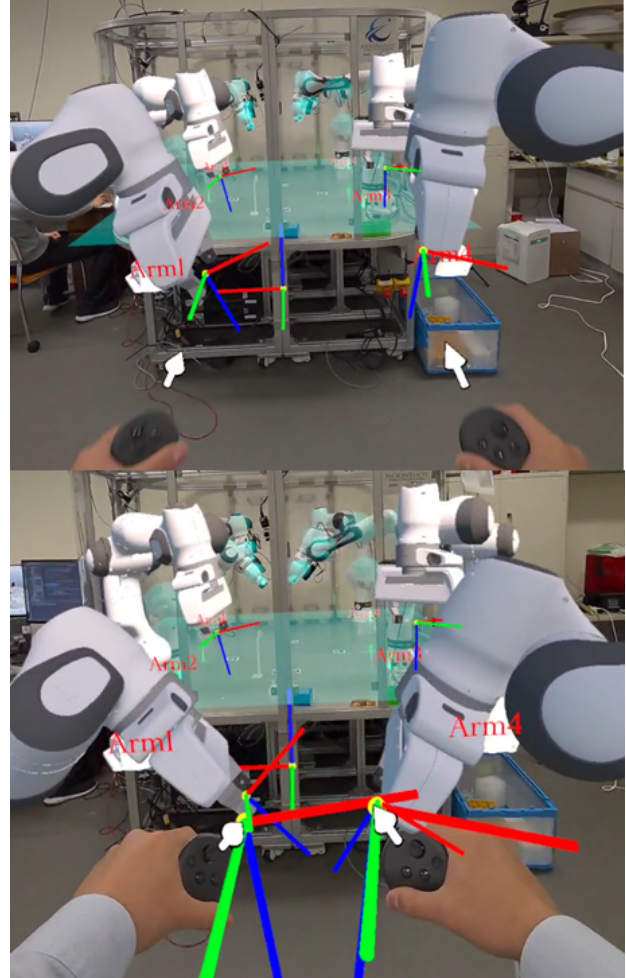


Figure 5: The subjective view of the user from the HMD running the user interface application: before starting the interactive operation (top) and during teleoperation (bottom).

particularly in multi-operator environments. These results represent a preliminary evaluation, highlighting the potential of the system while identifying areas for further refinement and testing. Use cases of the proposed MR human-robot interface include the manipulation of laboratory equipment, such as electronic pipettes, test tubes, beakers, Petri dishes, and other tools required for medium preparation under sterilized conditions via teleoperation. This intuitive human-robot interface enables the collection of demonstration data for training AI models to enhance robot autonomy.

4 CONCLUSION

The preliminary trials of the proposed MR human-robot interface demonstrate its potential for intuitive teleoperation of multiple robot arms by multiple operators. This approach departs from the traditional paradigm, where an operator is seated at a bi-manual leader controller interface and restricted to controlling two arms with left- and right-hand mappings. Instead, users can move freely, select a robot arm to control, and interactively operate it through one-to-one mapping of their hand motions to the robot's movements. Moreover, the proposed system supports multi-operator collaboration within the same scenario. By working under a unified reference coordinate frame, users can consistently observe each

other's actions as if the DT represented the real robotic system. This enables more intuitive and efficient collaborative teleoperation. However, the current implementation of the MR interface and DT does not account for the state of other objects in the scene that the robots interact with. Future efforts will focus on tracking these objects using AR markers, AI models for object pose estimation, or point cloud data to integrate their states into the DT and MR environments, thereby further bridging the gap between real and virtual environments.

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